METRIC

A-A-50620
28 January 1994

COMMERCIAL ITEM DESCRIPTION

ELECTRODE, pH, COMBINATION SEALED

The General Services Administration has authorized the use of this commercial item description as a replacement for DOD-E-24636 (SH) which is cancelled.

Scope: This commercial item description covers a combination pH and reference electrode which is used in conjunction with a digital indicating pH meter to measure the pH of boiler water aboard naval ships. The combination electrode and its connecting cable and connectors shall be compatible with the shipboard digital pH meter in use.

Salient characteristics:

The electrode shall be permanently sealed and non-refillable.

The outer body of the electrode shall consist of a rigid, impact-resistant opaque polymer material.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6630

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The outer body of the electrode shall extend below the bottom of the glass sensing element in order to prevent accidental breakage of the glass and to enable the electrode to be used as a stirring rod in glass or plastic beakers.

The electrode shall be shipped with the sensing and reference elements sealed in a liquid storage medium in order that the electrode be ready for use when received.

Electrode dimensions shall be from 13 to 18 centimeters in length, with an outside diameter from 1 to 2 centimeters.

The electrode electrical connecting cable shall be between 75 and 125 centimeters in length.

Each electrode electrical connecting cable shall be provided with connectors for inputting to a digital pH meter. The type of connectors to be provided shall consist of the following:

1. One U.S. standard type connector for the sensing element.
2. One pin-tip connector for the reference element.

In addition to the above connectors on the electrical connecting cable, a separate adaptor, U.S. standard electrode to BNC meter shall also be provided. This adaptor must be used if the digital pH meter in use requires a BNC input connector.

Performance characteristics shall be such that pH can be measured between 0.00 and 13.99 pH units from 0 degrees Celsius (°C) to 100 °C with a precision and accuracy of ± 0.05 pH units. For samples in equilibrium with the ambient temperature, the attainment of a stable pH reading shall require no longer than one minute from time of immersion of the electrode in the sample. For samples at temperatures other than ambient, no longer than five minutes shall be required for the pH determination. A meter reading that does not vary by more than ± 0.02 pH units in 30 seconds shall constitute a stable pH reading.

Workmanship: Electrodes shall be free from interior or exterior foreign material, smudges, scratches or loose particles which may interfere with normal operation of the electrode. Electrode surfaces shall be clean and free from any marks which would detract from a new appearance.

Regulatory requirements: The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent.

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Preservation, packaging, packing, labeling and marking: The preservation, packaging, packing, labeling and marking shall be as specified in the contract or purchase order. If unspecified, the preservation, packaging, packing, labeling and marking shall be consistent with acceptable commercial practice.

Notes: 1. Identification markings on the electrode shall show the name of the unit, date of manufacture, model number, serial number, national stock number and name of the contractor.

2. The contractor shall pack with each electrode, instructions for maintenance, use and storage of the electrode.

3. A combination sealed pH presently in the Federal Stock System has the following stock number: 9L 6630-01-063-9985.

4. pH sealed combination electrodes are commercial items available from:

Sensorex	11661 Seabord Circle Stanton, CA 90680 714-895-4344
Innovative Sensors	4745 East Bryson Street Anaheim, CA 92807 714-779-8781
Orion Research	529 Main Street Boston, MA 02129 617-242-3900

5. An adaptor for connecting a U.S. standard electrode to a BNC meter is available from:

Apple Scientific	8378 Mayfield Road Chesterland, OH 44026 216-729-3056 (Orion Cat. No. 090033)

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